Classtest-1. Digital Logic design. Time : 30 min.

Find the most appropriate answers(1-5)

1. According to the distributive law — 2.5
   a) A(B+C)=AB+AC (b) A(BC)=ABC (c) A+AB=A
2. The Boolean expression X=AB+CD represents,
   a) two ORs ANDed together (b) two ANDs ORed together (c) an exclusive OR.
3. According to the communicative law of addition,
   a) A=A+A (b) A+(B+C)=(A+B)+C (c) A+B=B+A.
4. According to DeMorgan's theorems, the following equality(s) are correct:
   a)(XYZ)'=X'+Y'+Z' (b) (A+B+C)'=A'B'C' (c) (AB)'=A'+B' (d) all of them
5. According to the associative law of multiplication,
   a) A+(B+C)=(A+B)+C (b) A(BC)=(AB)C (c) A+B=B+A
6. Map the following SOP expression on a K. Map and find the min. form of the expression: — 5
   B'C+AB'+AB'C+AB'CD'+A'B'CD+AB'C'D'.
7. Draw the truth table for the following SPOS: — 2.5
   X=(A'+B+C') (A+B'+C) (A'+B+C) (A+B+C)
8. Using Boolean algebra, simplify the following expression: — 4
   a) $\overline{A}BC + A\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} + A\overline{B}C + ABC$ (b) AB'+AB'C'+AB'C'D'+AB'C'D'E.
9. Apply DeMorgan's Law. — 2
   a) $\overline{A + B\overline{C} + D(E + \overline{F})}$ (b) $\overline{\overline{ABC} + D + F}$
10. Convert the following (a) Boolean expression into standard POS and (b) SSOP Expression to SPOS Expression. a) (A+B)(B+C) b) A'BC+ A'BC'+ AB'C+ ABC — 4

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